

estuarine influenced areas. Batangas, Cebu and Iloilo are located very near to protected areas and tourism areas. Batangas is within the center of the center of global marine biodiversity while Cebu is in the Mactan key biodiversity area. Manila has the highest number of foreign shipcalls while Cebu has the highest domestic shipcalls and second to Manila in international shipcalls.

PORT	SHIPCALLS	
	Foreign	Domestic
MANILA	2454	6,125
CEBU	1138	79,500
BATANGAS	958	13,196
SUBIC	313	136
CAGAYAN DE ORO	137	3,159
DAVAO	750	17,807
ILOILO	212	24,381
GENERAL SANTOS	112	704
ZAMBOANGA	40	41,27
LUCENA	74	4,428

Table 1. Top 10 ports in the Philippines in shipcalls (2020 data from PPA, CPA and SBMA)

The port of Manila has been documented to have a significant number of possible IAS. The on-going SAILS-PORTEC research program has detected IAS in Davao, Cebu and Matnog ports. These ports are adjacent to specific oil tanker pathways/routes. In Luzon where the refineries and oil storage facilities are located such as Batangas, are at higher risk. These loading ports are at high risk for IAS/MNIS and these are located near to international ports.

The shipcall statistics in Table 1 represent the year 2020, when the COVID 19 pandemic caused a global and domestic maritime transport slowdown. The average reduction in shipcalls is around 40%. Nonetheless, Manila and Cebu are likely the main ports that need to be closely monitored for potential IAS bioinvasion. In 2018, before the COVID-19 pandemic, Manila was experiencing port congestion with a report that ships may stay at berth for five days (Wallis, 2019). This will increase the risks for biofouling. Based on the 2021 statistics from the PPA, the average berthing time has been reduced to 1 day. This is a result of less shipping traffic due to the pandemic.